

Challenge 3: Impact Statement

Optimising RAG Document Retrieval for Agronomic Advice

Smallholder farmers, agricultural advisers, and extension workers often need timely information about crop diseases, pests, nutrient deficiencies, soil management, fertiliser use, and climate adaptation.

However, relevant agricultural knowledge is distributed across extension factsheets and technical resources, while farmer questions may use everyday language that differs from terminology in those documents. This project aims to improve the retrieval and ranking of relevant agricultural extension documents for farmers' questions, helping downstream RAG systems ground responses in more appropriate evidence. The potential positive impact is improved access to relevant agricultural knowledge. Farmers and extension workers could spend less time searching large collections and have a better chance of finding documents that directly address a farming problem. Agricultural organisations could also benefit from more efficient knowledge access. In the longer term, better retrieval could support more informed decisions about crops, pests, diseases, soils, fertilisers, and climate adaptation, although this project does not itself demonstrate improved farm outcomes. The project also carries risks. An irrelevant, outdated, or poorly ranked document could contribute to misleading advice in a downstream RAG system. The dataset is English-only and has finite, uneven coverage, so it may not represent farmers who primarily use Hausa, Yoruba, Igbo, Nigerian Pidgin, or other languages. It is also a fixed competition dataset rather than a community-validated resource.

These limitations create risks of exclusion, overgeneralisation, and misplaced confidence. Our solution will be guided by accessibility, fairness, inclusivity, accuracy, reliability, transparency, and community well-being. We will evaluate retrieval using graded relevance and nDCG@5, compare results against the TF-IDF baseline, test hard-negative cases, and preserve document identifiers so retrieved evidence can be traced and checked. We will clearly communicate that improved retrieval does not guarantee correct agronomic advice and that further expert and community validation is needed before real-world deployment. The impact assessment remains uncertain because the competition dataset and hidden test judgments do not fully represent real

farming contexts. Future work should expand multilingual and locally relevant resources and involve farmers and agricultural experts in validation. Overall, the project seeks to make agricultural knowledge retrieval more relevant, transparent, and useful while avoiding claims that exceed the evidence.

Reflection 1: Most Significant Positive Impact

The most significant positive impact is helping smallholder farmers and agricultural extension workers find relevant agricultural knowledge more quickly. Better access to information about crop diseases, pests, soils, fertilisers, and climate adaptation can support more informed decisions and make agricultural advisory services more efficient. This is important because timely, relevant information can be valuable when farmers respond to problems affecting crop productivity and livelihoods.

Reflection 2: Most Significant Risk and Mitigation

The most significant risk is that an incorrectly ranked or outdated document could contribute to misleading downstream agronomic advice. We will manage this through nDCG@5 evaluation, graded relevance judgments, comparison with the TF-IDF baseline, hard-negative testing, traceable document evidence, and clear communication that the system is a retrieval research prototype rather than a substitute for agricultural expertise.

Reflection 3: Responsible AI Learning

Completing the Problem Statement, Data Card, and Impact Statement has shown the team that responsible AI is not only about technical performance. The Data Card exposed limitations in language, geographic coverage, labeling, and community validation, while the Impact Statement required us to consider who may benefit, who may be excluded, and what could go wrong. We now see evaluation, transparency, inclusivity, and honest communication of limitations as part of system design rather than activities added after development.